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HFA POSITION PAPER

Right heart dysfunction and failure in heart failure with preserved ejection fraction: mechanisms and management. Position statement on behalf of the Heart Failure Association of the European Society of Cardiology

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There is an unmet need for effective treatment strategies to reduce morbidity and mortality in patients with heart failure with preserved ejection fraction (HFpEF). Until recently, attention in patients with HFpEF was almost exclusively focused on the left side. However, it is now increasingly recognized that right heart dysfunction is common and contributes importantly to poor prognosis in HFpEF. More insights

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into the development of right heart dysfunction in HFpEF may aid to our knowledge about this complex disease and may eventually lead to better treatments to improve outcomes in these patients. In this position paper from the Heart Failure Association of the European Society of Cardiology, the Committee on Heart Failure with Preserved Ejection Fraction reviews the prevalence, diagnosis, and pathophysiology of right heart dysfunction and failure in patients with HFpEF. Finally, potential treatment strategies, important knowledge gaps and future directions regarding the right side in HFpEF are discussed.

Keywords Heart failure • Right ventricular function • Pulmonary hypertension • Biomarkers • Heart failure with preserved ejection fraction

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Introduction

For many years, the right ventricle (RV) was largely neglected in the consideration of left-sided heart failure. In the last two decades, however, several studies have clearly demonstrated that right ventricular dysfunction (RVD) is not only common in left heart failure but its presence also strongly contributes to increased morbidity and mortality.[1] The mechanisms behind the development of RVD in patients with left heart failure have been well established, but these data were, until recently, almost exclusively obtained in patients suffering from heart failure with reduced ejection fraction (HFrEF).[2]

It is now evident that RVD is highly prevalent and contributes to poor prognosis in patients with left-sided heart failure with preserved ejection fraction (HFpEF).[3–5] However, the underlying pathways that lead to RVD in HFpEF are far less clear. In addition, patients with idiopathic pulmonary arterial hypertension (PAH) and chronic obstructive pulmonary disease (COPD) also present with signs and symptoms of heart failure with a preserved left ventricular (LV) ejection fraction. Better insight into the mechanisms causing the development of RVD and its clinical role in HFpEF may aid to (i) our understanding of this complex disease, and (ii) develop novel treatment strategies to improve outcomes. The present position paper reviews the pathophysiology, potential treatment strategies, knowledge gaps, and future directions regarding the right side of the heart in HFpEF.

Prevalence and prognosis

Puwanant *et al.*[6] were one of the first that investigated RV systolic dysfunction in HFpEF, demonstrating a prevalence of RVD of 33%, 40%, and 50%, as categorized by reduced RV fractional area change (FAC), tricuspid annular plane systolic excursion (TAPSE) and tricuspid annular systolic velocity (RV S'), respectively. Reporting exact prevalence data for RVD is challenging, because HFpEF severity across different studies has varied, as well as the echocardiographic methods, criteria and cut-off values employed for the assessment of RVD.[3,4,7–9] Some of these cut-off values for RVD are less well established or have also changed over time.[10–12] Aschauer *et al.*[8] used cardiac magnetic resonance (CMR) imaging, and RVD was present in 19% of the patients.

It is important to keep in mind that the prevalence of RVD may also be influenced by inclusion criteria that are applied to define HFpEF. The classic signs of left- and right-sided heart failure are not mutually exclusive. For instance, the often used Framingham criteria for the diagnosis of HFpEF include signs consistent with right-sided decompensation, such as hepatomegaly and peripheral oedema. Furthermore, the inclusion of patients with (concomitant) pulmonary vascular disease (PVD) caused by mechanisms other than left heart failure (e.g. COPD and PAH) may result in the over-representation of RVD unrelated to left-sided heart failure.[13,14] Furthermore, the application of different exclusion criteria among individual studies may also affect the prevalence rate of RVD. For instance, renal dysfunction is strongly associated with pulmonary hypertension (PH) and RVD.[15] The prevalence of RVD may therefore be relatively higher in community-based studies compared to clinical trials, because (severe) renal dysfunction is often an exclusion criterion in the latter. Because signs suggestive of RVD are used to diagnose heart failure in community studies, there may be a relatively higher identification of HFpEF patients with RVD compared to HFpEF patients without RVD in these studies compared to clinical trials, in which more often stringent HFpEF criteria are used.

Finally, it is challenging to define the prevalence of RVD considering the small number of patients among individual studies. In a recent, large meta-analysis, the prevalence of RVD was 18%, 28%, and 21% using RV FAC, TAPSE and RV S', respectively.[5] Thus, despite variable reports, methods and criteria, the best available current data indicate that RVD is present in *at least* one-fifth and *potentially* up to 30–50% of patients with HFpEF.

Most cardiovascular deaths in HFpEF can be attributed to end-stage circulatory failure.[16] It is now increasingly evident that RVD is also a major predictor of the clinical course of HFpEF.[5] Melenovsky *et al.*[3] reported a 2.2-fold increased risk for all-cause mortality per 7% decrease in FAC after adjustment for pulmonary pressures. In the study by Aschauer *et al.*[8], the hazard ratio for outcome was 4.9 for patients with RV ejection fraction <45% on CMR. Pooled data from individual studies recently showed that the risk of mortality increases with ~26% per 5 mm decrease in TAPSE and with ~16% per 5% decrease in FAC.[5] In a recent prospective study, 55% of patients with HFpEF died with clinical signs of right heart failure.[17]

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Diagnosis

Right heart dysfunction vs. failure

The distinction between RVD and right heart failure may be likened to that between LV systolic dysfunction and left heart failure, in that the former is defined by abnormal values of functional parameters, whereas the latter is defined by haemodynamic decompensation with typical clinical signs and symptoms.

Right heart *dysfunction* is present when a measure of RV function falls outside the recommended range of normal (*Table 1*).[11] In the absence of clear normal values, the range that is published varies and hinges on agreement to what defines normal and abnormal. Right ventricular dysfunction may be asymptomatic, but there may be evidence of adaptive RV remodelling in response to increased afterload such as RV hypertrophy, which seems present in ~22% of patients with HFpEF and ~45% of patients with PH-HFpEF.[18] Preliminary evidence suggests that there may be increased reliance on a contraction pattern more suited for increased afterload to preserve global systolic function (e.g. reduced longitudinal shortening but enhanced transverse contraction), analogue to the LV.[19] However, opposite results were observed in a prior echocardiographic study, demonstrating that reduced transverse RV function is rather a reliable indicator of the presence of high pulmonary artery (PA) pressure, whereas reduced RV longitudinal function was associated with overall impairment of cardiac function.[20] This implies that impaired RV transverse function may be an early indicator of the presence of high pulmonary pressures, which is of interest in the *screening* for PH. On the other hand, reduced RV longitudinal function in the presence of high pulmonary pressures yields an important role in the *follow-up* of patients with known PH. Given these exemplary observations, multiple independent measurements that reflect RV systolic function are preferably used and we herein propose that RV systolic dysfunction is present in HFpEF if (i) at least two echocardiographic parameters for RV systolic function are below their recommended cut-off value (*Table 1*), or (ii) if RV ejection fraction measured with CMR is <45%. Still, it must be noted that the majority of data regarding RVD are obtained from patients with HFrEF instead of HFpEF.

Furthermore, while there remain large data available on cut-off values for RV dysfunction, from a clinical point of view it is also important to know which cut-off values of individual functional parameters are associated with poor prognosis in HFpEF. In general, these cut-off values are lower than the ones used to only define dysfunction. For instance, Ghio *et al.*[9,21] identified TAPSE <14 mm to be strongly associated with adverse prognosis, both in HFrEF and HFpEF. However, data in this regard are scarce in HFpEF and reported values also vary widely. Additional prospective studies are needed to identify clear RVD cut-off values associated with prognosis that have clinical relevance in HFpEF.

Right heart *failure* is a clinical diagnosis with signs and symptoms of systemic congestion in combination with structural and/or functional abnormalities of the right heart.[22] Right heart failure may be caused by RVD itself or RV remodelling, including annular dilatation causing tricuspid insufficiency and right atrial dysfunction. These changes ultimately cause symptoms of exertional dyspnoea and reduced exercise capacity and/or signs of right-sided decompensation such as jugular venous distension, hepatomegaly, ascites, and peripheral oedema (*Table 1*).[22] Ultimately, after long-standing right-sided pressure and/or volume overload, the right heart is unable to provide adequate blood flow through the pulmonary circulation at a normal central venous pressure and cardiac output decreases, which ultimately leads to death, if not treated adequately.[22]

It is important to acknowledge that staging phases of RVD and right heart failure vary of time and some patients may not have RVD at rest, but rather during exercise. In contrast to left-sided heart failure, there is currently no clear staging of right heart failure, although attempts have been made to develop a staging system.[2] For right heart failure in the setting of HFpEF, we herein propose a staging system of RVD and right heart failure (*Table 2*). However, the proposed staging system needs further validation in a prospective setting.

Echocardiography

Right ventricular function can be assessed using several methods and echocardiography is generally the first-line tool (*Figure 1*). TAPSE is most frequently used and has an independent prognostic value in HFpEF.[4,8,23–25] The currently recommended lower limit cut-off for TAPSE is <17 mm,[12] although prior studies frequently used <16 mm. Fractional area change (with a lower limit of normal <35%[12]) is also commonly used, and is predictive of all-cause mortality and heart failure hospitalizations in HFpEF.[3,8,24,26]

Because for TAPSE and FAC there is the most available evidence in relation to prognosis, these two measures should preferably be assessed, if possible, in all patients with HFpEF. Other echocardiographic indices related to RVD used in HFpEF include RV S',[3,6,7,27–29] and RV longitudinal strain (*Table 1*, *Figure 1*).[7,29] The latter has advantages because it is angle-independent and has the potential ability to detect subtle, regional myocardial changes in HFpEF, when conventional echocardiographic parameters remain within the normal range.[29] In addition, TAPSE and RV S' are sometimes falsely elevated when the LV is hyper-dynamic, due to tethering effects of the LV secondary to both ventricles sharing myocardial muscle fibres.[30] Right ventricular strain may be lesser influenced by these tethering effects in this context. The association between RV strain and prognosis has been described in patients with HFpEF,[26] but further evidence from other prospective HFpEF cohorts, using dedicated RV-focused strain imaging, is needed.

The RV index of myocardial performance (RIMP) is an index of global RV function. Higher values of RIMP indicate impaired myocardial performance (*Table 1*, *Figure 1*).[12] Furthermore, 3D echocardiography is increasingly being used and RV ejection fraction <45% is recommended as cut-off value for RVD (*Table 1*, *Figure 1*),[12] yet data on RV ejection fraction assessed with 3D echocardiography in HFpEF are scarce.

Assessment of pulsed-wave Doppler for tricuspid inflow (i.e. tricuspid E/A ratio) or hepatic vein diastolic flow, tissue Doppler for lateral tricuspid annular diastolic velocity (e'), and right atrial size, may all be useful to measure RV diastolic function.[31] However, these diastolic indices are influenced by age, respiration, heart rate, pulmonary pressures, and other covariates.[31] Right ventricular

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Table 1 Right heart dysfunction and failure

Right-sided structural and functional abnormalities on echocardiography	
Presence of right ventricular systolic dysfunction	
• TAPSE	<17 mm
• RV FAC	<35%
• RV S'	<9.5 cm/s
• RV FWLS	> −20% (<20 in magnitude with the negative sign)
• RIMP (pulsed Doppler)	>0.43 (pulsed Doppler) or >0.54 (tissue Doppler)
• 3D RV ejection fraction	<45%
Signs of right-sided pressure/volume overload	
• Tricuspid regurgitation	
• RV basal end-diastolic diameter	>41 mm
• RVEDD/LVEDD	>1.0
• Septal shift or D-shaped LV in systole and/or diastole	
• RV wall thickness	>5 mm
• Inferior vena cava diameter	>21 mm
• Inferior vena cava collapsibility	<50%
• Tricuspid regurgitation peak systolic velocity	>2.8 m/s
• Right atrial end-systolic area	>18 cm^2
Clinical evidence of right heart failure	
• Evidence of right-sided structural and/or functional abnormalities (see above), in combination with:	
Symptoms, e.g.:	
• Exertional dyspnoea	
• Fatigue	
• Dizziness	
• Ankle swelling	
• Epigastric fullness	
• Right upper abdominal discomfort	
Signs, e.g.:	
• Jugular venous distension	
• Peripheral oedema	
• Hepatomegaly	
• Ascites	
• Third heart sound	

FAC, fractional area change; FWLS, free wall longitudinal strain; LV, left ventricular; LVEDD, left ventricular end-diastolic diameter; RIMP, right ventricular index of myocardial performance; RV, right ventricle; RVEDD, right ventricular end-diastolic diameter; RV S', systolic velocity of the lateral tricuspid valve annulus; TAPSE, tricuspid annular plane systolic excursion.
Echocardiographic cut-off values adapted from the ASE/ESC guidelines.[11,12,99]

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Table 2 Staging of right heart dysfunction and failure in heart failure with preserved ejection fraction

Stage 1	At risk for right heart failure without RV dysfunction and without signs/symptoms of right heart failure.
Stage 2	RV dysfunction without signs/symptoms of right heart failure.
Stage 3	RV dysfunction with prior or current signs/symptoms of right heart failure.
Stage 4	RV dysfunction with refractory signs/symptoms of right heart failure at rest (requiring specialized interventions).

RV, right ventricle.
Adapted from Haddad *et al.*[2]

diastolic function is not routinely assessed in clinical practice but dysfunction can be present, at least during exercise, early in the course of HFpEF and in parallel with LV diastolic dysfunction, probably due to combined myocardial processes that affect both ventricles simultaneously.[32]

Of note, a correct and complete assessment of LV diastolic function (E/A, e', E/e' and left atrial size) is important to diagnose RVD in the setting of HFpEF, because in some cases there is a discrepancy between the severity of LV diastolic dysfunction and RVD.

Echocardiography is also the first-line tool for the evaluation of increased right-sided pressures.[11] Given the high prevalence of PH in HFpEF, it is recommended to asses direct and indirect echocardiographic signs related to increased pulmonary pressures in all patients with HFpEF. This primarily includes the estimation of PA systolic pressure (PASP in mmHg) using the tricuspid regurgitation (TR) jet velocity and estimated central venous pressure. Other useful indices include RV basal end-diastolic diameter (>41 mm), RV hypertrophy (wall thickness >5 mm), right atrial dilatation (right atrial end-systolic area >18 cm^2), and inferior vena cava size and collapsibility for the estimation of right atrial pressure.[11] More specific measures that can be obtained for a non-invasive estimation of pulmonary vascular resistance (PVR) include RV outflow tract (RVOT) notching on the pulsed-wave Doppler profiles, and the peak TR velocity/RVOT velocity–time integral.[33] For the latter, a value <0.18 suggests elevated PVR to be unlikely.[33] Echocardiography can also be used to differentiate between PAH and PVD-HFpEF, using left-to-right atrial size ratio and interatrial and interventricular septal bowing.[33] In patients with PH-HFpEF, interventricular septal bowing may be less pronounced than in PAH due to typically higher left-sided systolic pressures in HFpEF compared to patients with PAH.

Cardiac magnetic resonance imaging

The assessment of the right heart using echocardiography may be challenging, due to its complex geometry, limited echocardiographic windows and high prevalence of obesity and COPD in HFpEF. Therefore, CMR is increasingly being used and is currently recommended (class IC) in patients with suspected or established heart failure with poor acoustic window, for the assessment of myocardial structure and function (taken into account for cautions/contra-indications to CMR).[34] Reduced RV ejection fraction on CMR is associated with worse prognosis in HFpEF,[8,35] and <45% is most commonly used as cut-off for RVD.[36] CMR is also reliable in measuring RV volume and hypertrophy, which may be useful in the setting of PVD-HFpEF. In addition, the PA to aorta diameter ratio seems useful as non-invasive indicator of the presence of PH in HFpEF.[37]

Quantification of focal and diffuse myocardial fibrosis is also feasible by CMR, using late gadolinium enhancement, T1 time, and extracellular matrix fraction.[37,38] Lower LV myocardial post-contrast T1 time is associated with higher extracellular matrix and collagen content, and predicts poor prognosis in HFpEF.[38] Late gadolinium enhancement can typically be observed near the septal insertion points of the RV, and this finding is reported in patients with PH[39] and hypertrophic cardiomyopathy,[40] as well as in patients with various aetiologies of HFpEF.[8] As for the RV itself, the myocardial wall is often too thin for reliable quantitative assessment of T1 time using the standard modified look–locker sequences. However, high-resolution look–locker images have higher pixel density and may be reliable for the estimation of extracellular volume content in the thin-walled RV myocardium.[41]

Right ventricular–pulmonary artery coupling

The RV is exquisitely sensitive to changes in afterload,[42] and this heightened afterload-dependence is exaggerated in patients with HFpEF.[3] In view of the frequent presence of PVD and increased RV afterload in HFpEF, consideration of RV–PA coupling is important.

The gold standard assessment of RV–PA coupling involves invasive pressure–volume (PV) loops. In normal conditions, the PV loop of the RV is triangular shaped, due to lower resistance and higher compliance of the pulmonary vasculature, as compared to the systemic circulation.[42] In response to increased vascular load, the RV adapts with a 4- to 5-fold increase in myocardial contractility, haemodynamically visualized by a more rectangular 'LV-shaped' PV loop and a leftward shift of the end-systolic PV relationship ('coupling').[42] In the next progressive stage, the RV becomes more dilated and the end-systolic PV relationship shifts to the right, resulting in lower myocardial contractility ('uncoupling'). Heart rate increases to maintain sufficient cardiac output, which will lead to increased wall stress and oxygen demand.[42] Ultimately, after longstanding high metabolic demand, the RV is unable to maintain cardiac output, which is accompanied by signs and symptoms of severe right heart failure.[42] The attention for the importance of RV–PA coupling in left-sided heart failure has been increasing, because conventional systemic vasodilators used to treat left-sided heart failure were observed to have no beneficial effect on the right heart. However, invasive PV assessments

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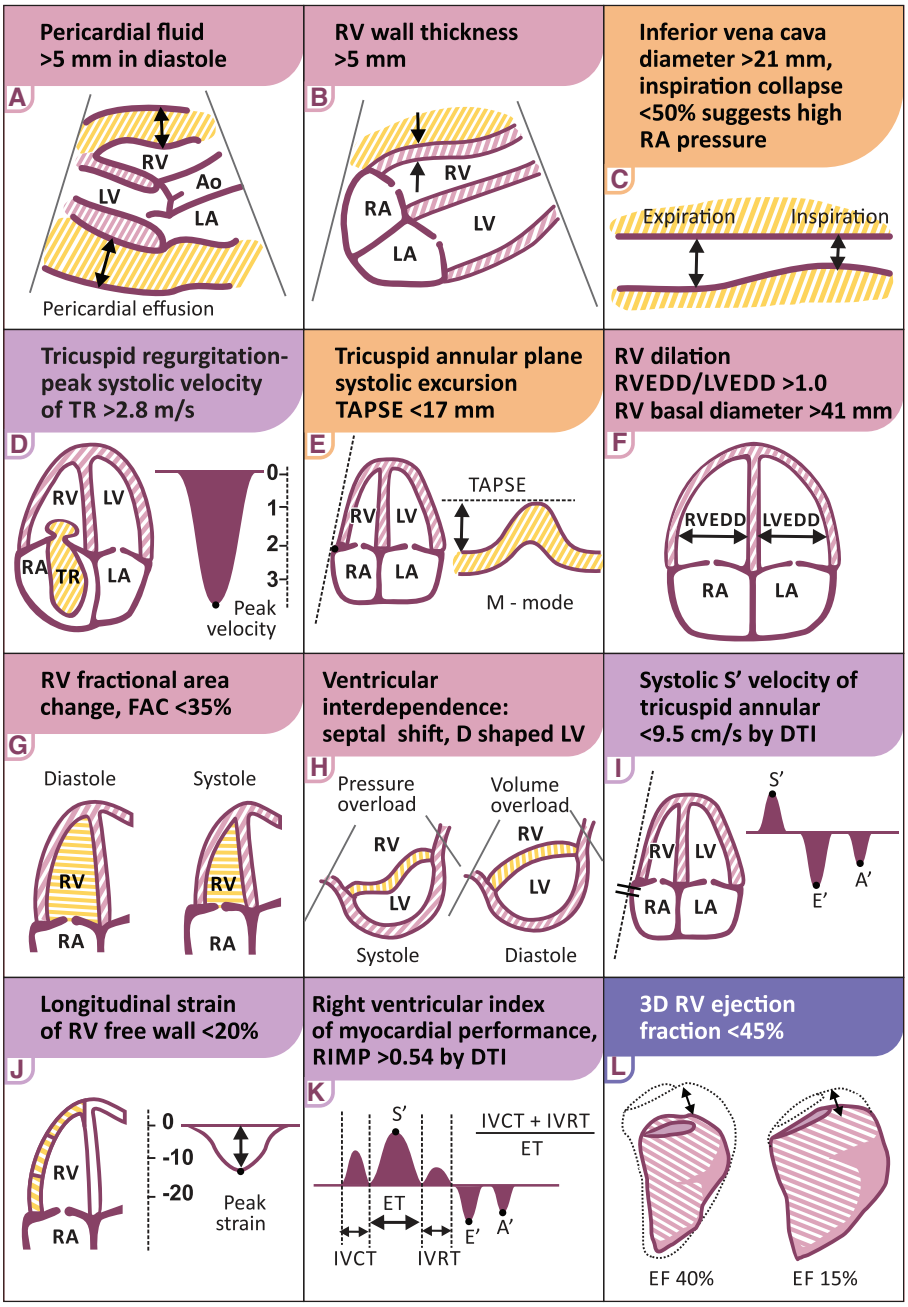


Figure 1 Graphic representation of echocardiographic parameters in the assessment of right ventricular dysfunction. Ao, aorta; DTI, Doppler tissue imaging; EF, ejection fraction; ET, ejection time; FAC, fractional area change; IVC, inferior vena cava; IVCT, isovolumic contraction time; IVRT, isovolumic relaxation time; LA, left atrium; LV, left ventricle; LVEDD, left ventricular end-diastolic diameter; RA, right atrium; RIMP, right ventricular index of myocardial performance; RV, right ventricle; RVEDD, right ventricular end-diastolic diameter; TAPSE, tricuspid annular plane systolic excursion; TR, tricuspid regurgitation. Reproduced from Harjola *et al.*[99] with permission of the authors. Copyright © 2016.

are rather cumbersome and not without risk.[42] Recent studies have examined RV–PA coupling non-invasively in patients with PH.[43–45] Guazzi *et al.*[46] proposed an echocardiographic estimate of RV–PA coupling in heart failure, using both TAPSE and PASP. Reduced TAPSE/PASP ratio was associated with worse prognosis, both in HFrEF and HFpEF,[9,46–48] and a prognostic cut-off of this ratio was identified: <0.36.[9,46,48] In a study with parallel invasive pulmonary haemodynamic assessment, the TAPSE/PASP ratio was gradually reduced across the PA resistance–compliance hyperbolic relationship, which suggests that this simple, non-invasive marker could reflect the 'operating load' the RV has to work against and this ratio may already be reduced when cardiac output is still preserved.[47]

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Reduced TAPSE/PASP ratio also seems to be a reliable non-invasive parameter to identify HFpEF patients with a high likelihood of having additional pre-capillary PH.[48]

Finally, RV stroke work index can be estimated non-invasively in patients with heart failure, by calculating the RV contraction pressure index as the product of TAPSE and the transtricuspid systolic pressure gradient.[49]

Right ventricular function during stress testing

Evaluation of right-sided filling pressures, RV–PA coupling and RV systolic and diastolic function with stress testing such as exercise and preload augmentation has the potential to enhance haemodynamic assessment at different stages of HFpEF.[32,50] Patients with advanced HFpEF and concomitant PVD may have normal RV functions at rest, but RVD may be present during exercise, if the ability of the pulmonary vasculature to dilate in response to increased volume load is lost. Right ventricular strain may be a sensitive indicator for this phenomenon.[50] On the contrary, impaired RV function at rest may not necessarily imply a worse adaptive contractile response during exercise in patients with heart failure.[51] An additive role for cardiopulmonary exercise testing in patients with PAH is recently highlighted in the 2015 European Society of Cardiology guidelines on the diagnosis and management of PH,[52] and may also be useful to explore the various phenotypes and different levels of risk in patients with heart failure.[51,53]

Remarkably, studies by Borlaug *et al.*[32] have shown that RV–PA uncoupling occurs during exercise, even in the earlier stages of HFpEF. Elevation in left atrial pressure and impaired left atrial strain response during exercise (despite preserved LV ejection fraction) is pathognomonic for HFpEF,[54,55] and this acutely shifts the PA resistance–compliance relationship leftward, causing an increase in pulsatile RV load.[55,56] Assessment of PA pressure–flow relationships with exercise has been proposed as a useful and prognostically relevant metric to evaluate vascular reserve in HFpEF,[57] and some HFpEF phenotypes, such as those with obesity-related HFpEF, may display greater abnormalities in RV–PA coupling during exercise that potentially move forward towards therapeutic implications.[58]

Aetiology

Pulmonary hypertension

The most important mechanism of RVD in HFpEF is contractile impairment and afterload mismatch in the setting of PH-HFpEF (*Figure 2*).[3,4] Left ventricular diastolic dysfunction and loss of left atrial compliance in the setting of HFpEF impose a pulsatile load on the pulmonary venous system leading to a 'passive' increase in pulmonary pressure.[59] Clinical features that are suggestive of HFpEF in patients with suspected PH include older age, typical co-morbidities such as atrial fibrillation (AF) and obesity, and left atrial dilatation.[60] An additional component of pulmonary vasoconstriction with or without pulmonary vascular remodelling, may lead to further increases in pulmonary pressure that are not in proportion to left-sided filling pressures in HFpEF.[61] In specialized centres, right heart catheterization may be recommended to distinguish between PAH and HFpEF-PH and/or to confirm HFpEF if the diagnosis is uncertain.[60] Simple haemodynamic equations obtained from right heart catheterization, such as PVR, transpulmonary gradient and diastolic pressure gradient (DPG), can distinguish the different haemodynamic conditions.[59] Current consensus classifies PH with left heart disease as 'isolated post-capillary PH' [pulmonary capillary wedge pressure (PCWP) >15 mmHg, mean PA pressure ≥25 mmHg with PVR ≤3.0 WU and/or DPG <7 mmHg] and 'combined post- and pre-capillary PH' (PCWP >15 mmHg, mean PA pressure ≥25 mmHg with PVR >3.0 WU and/or DPG ≥7 mmHg).[52] An accurate assessment of PCWP is essential, requires great attention to detail and should be measured at end-expiration and mid A-wave. In many patients with HFpEF, PCWP might be <15 mmHg at rest, in fasting state and with optimal diuretic therapy, and PCWP may typically rise with exercise or after a fluid challenge, which enhances the diagnosis of HFpEF.[52,54] Consequently, PCWP <15 mmHg at rest in combination with increased PA pressure does not necessarily implies isolated pre-capillary PH.[52,54] Standardized diagnostic protocols should be designed specifically for these occurrences. In patients with high PVR indicating combined post- and pre-capillary PH, vasoreactivity testing may be useful to identify a reactive pulmonary arterial component.[60] Individual haemodynamic characteristics obtained with right heart catheterization provide more patient-tailored strategies or participation in clinical trials.

Pulmonary hypertension is a common finding in patients with HFpEF, and is associated with worse symptoms, reduced exercise capacity, higher natriuretic peptide levels, and increased hospitalization rates and mortality.[48,62,63]

In a population-based study from Lam *et al.*[64], up to 83% of all-comers with HFpEF had non-invasive evidence of PH. In a later invasive study by Melenovsky *et al.*[3], 81% of HFpEF patients had PH. Data from the large TOPCAT clinical trial revealed a much lower PH prevalence rate of 36%.[65] The presence of PH may thus vary considerably among different HFpEF populations and may be influenced by the different stages of HFpEF severity enrolled in the different studies (e.g. clinical trials vs. community-based studies), as well as the frequent exclusion of co-morbidities in most clinical trials that are strongly associated with PH, such as severe renal dysfunction and significant pulmonary disease.[5,66] If pulmonary haemodynamics are not invasively assessed, it remains unclear which amount of included patients with HFpEF have PH due to (concomitant) PAH, COPD, or other systemic diseases.

Ventricular interdependence

One of the key features in HFpEF is LV diastolic dysfunction. Although, global LV ejection fraction is by definition preserved in HFpEF, there is evidence of LV contractile dysfunction.[67] Because both ventricles share myocardial muscle fibres and the interventricular septum, approximately 20 to 40% of RV systolic performance can be attributed to LV contraction.[30] This concept of systolic ventricular interdependence might be the mechanism behind the previously reported associations between reduced ejection fraction and the higher prevalence of RVD in HFpEF.[3,4,8] In the setting

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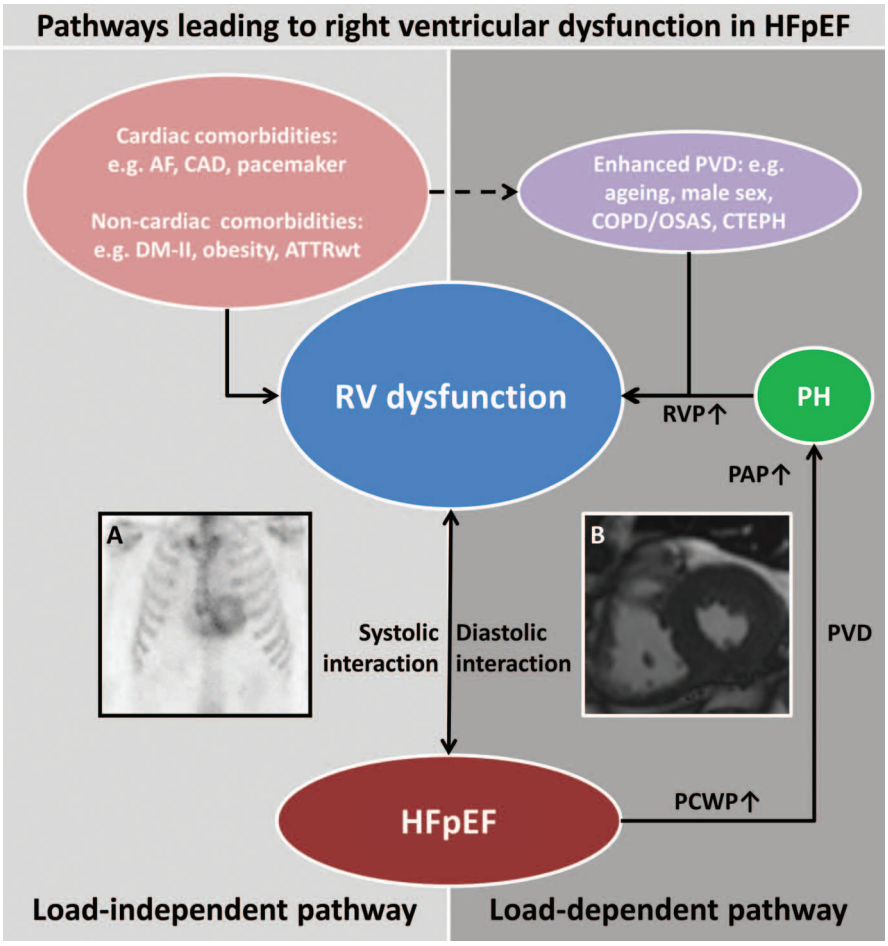


Figure 2 Pathways leading to right ventricular dysfunction (RVD) in heart failure with preserved ejection fraction (HFpEF). The predominant mechanism for the development of RVD is the load-dependent pathway, caused by (i) post-capillary pulmonary hypertension (PH) due to increased left-sided filling pressures, and (ii) aggravated pulmonary vascular disease (PVD) with chronic pulmonary congestion and concomitant factors such as ageing, male sex, chronic obstructive pulmonary disease (COPD), and chronic thromboembolic pulmonary hypertension (CTEPH). Chronically increased afterload in PVD-HFpEF will lead to right ventricular (RV) hypertrophy, dilatation and dysfunction (*B*). Increased right-sided pressure will also lead to leftward septal bowing and this 'diastolic ventricular interaction' further impairs left-sided filling. In isolated post-capillary PH, left atrial size typically increases before or together with increase in pulmonary artery pressure (PAP) while in the setting of additional pre-capillary PH, right atrial size may surpass left atrial size. There are also several load-independent factors associated with development of RVD. These include: (i) mechanistic factors, such as atrial fibrillation (AF), right-sided pacing and 'systolic ventricular interaction', and (ii) intrinsic myocardial processes that involve systemic inflammation and endothelial dysfunction caused by non-cardiac co-morbidities and for instance right-sided involvement in wild-type transthyretin amyloidosis (ATTRwt) (*A*). CAD coronary artery disease; DM-II, type 2 diabetes mellitus; OSAS, obstructive sleep apnoea syndrome; PCWP, pulmonary capillary wedge pressure; RVP, right ventricular pressure.

of regional LV contractile dysfunction despite preserved global LV ejection fraction in HFpEF, the direct contribution of LV contraction to RV systolic function is also reduced. This left-to-right ventricular contribution is also lost in the setting of RV pacing in HFpEF.

Diastolic ventricular interdependence (DVI) is also important in HFpEF. Approximately 30 to 40% of LV diastolic pressure is related to extrinsic forces including right heart pressure and pericardial restraint.[68] Even a slight increase in pulmonary pressure already leads to a leftward septal shift and impaired LV diastolic compliance induced by DVI in the absence of prior intrinsic LV disease.[69] This effect is already present in patients with even mild to moderate PH, and may aggravate clinical manifestation of (exertional) dyspnoea.[69] DVI with inhomogeneous septal movement can be assessed using 2D speckle tracking echocardiographic strain (Supplementary material online, *Figure S1*). DVI might also belong to the mechanisms that contribute to septal mechanical delay and asynchronicity, which can be found in ~20% of the patients with HFpEF.[70] Haemodynamic changes induced by DVI are also accompanied by neurohormonal activation.[69] Recent data have shown that DVI is increased in HFpEF patients with obesity, coupled with increases in right heart remodelling and dysfunction.[58]

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Intriguingly, the degree of DVI is synergistically enhanced as PA pressure increased in these patients. An important repercussion of this phenomenon is that higher pulmonary venous pressure is required to achieve a given LV transmural distending pressure, which is then transmitted back to further increase RV afterload.[58]

Atrial fibrillation

Atrial fibrillation is also common in HFpEF.[71] Several studies have indicated a link between AF and the presence of RVD in HFpEF.[3,4,9,72] Both may relate to HFpEF, in which increased LV filling pressures leads to left atrial stretch and remodelling on the one hand,[71] and to increased pulmonary pressures and RV afterload on the other. Indeed, HFpEF patients with AF have higher PCWP and mean PA pressure, compared to HFpEF patients in sinus rhythm.[72]

Left atrial function and adverse remodelling are important in the context of heart failure.[73] As left atrial function deteriorates, particularly with development of AF, there is more profound pulmonary vascular dysfunction and consequent impairment in RV function in patients with HFpEF.[74] However, it is not just afterload mismatch, as RV shortening is worse in HFpEF patients with AF for any given PA pressure load.[3] Possible mechanisms might be related to decreased RV longitudinal contraction and rhythm irregularity with negative inotropic effects in the setting of AF,[75,76] although we lack prospective studies that addressed such hypotheses in HFpEF. Conversely, RVD is also a strong predictor of the occurrence of AF after acute decompensated heart failure.[77]

Atrial fibrillation is also prevalent in patients with PH that is not caused by left heart failure (~20 to 23%),[78,79] and the presence of PH is associated with more right atrial dilatation and higher right atrial pressures compared to PH patients without AF.[79] Reduced TAPSE and right atrial dilatation were previously linked to both history and development of AF in patients with hypertrophic cardiomyopathy.[80] To what extent *right atrial* overload beyond *left atrial* overload may facilitate the onset or progression of AF, in patients with PVD-HFpEF, needs to be evaluated.

Coronary artery disease

Coronary artery disease is associated with more advanced impairment of RV function in both HFpEF and HFrEF.[3,4,9] Although patients with large myocardial infarctions do not typically present as HFpEF, coronary artery disease is also common in HFpEF, up to 50–66%.[81] While, the RV has the ability to recover from an acute ischaemic insult and isolated RV infarction is rare,[82,83] single occlusion of proximal right coronary artery or of dominant circumflex coronary artery may result in RV systolic dysfunction while global LV systolic function is preserved. Especially in the setting of RV hypertrophy and PH, coronary artery disease might enhance the mismatch between perfusion and demand.

Furthermore, approximately one-third of HFpEF patients with coronary artery disease had previously undergone coronary artery bypass grafting.[81] In patients undergoing such major cardiac surgery and/or valve replacement, TAPSE may be considerably reduced, even up to 1 year after surgery.[84] Reduction in RV filling and contraction after cardiac surgery seems independent of cardiopulmonary bypass and is not seen in patients receiving percutaneous interventions.[85] The significant reduction in TAPSE may therefore be mediated by pericardial adhesions following pericardiectomy.[85] Potential other mechanisms have also been described, such as cytokine release, RV infarction due to ischaemia or air emboli, and inflammation or effusion post-surgery.[85]

Besides the contribution of obstructive epicardial coronary artery disease, coronary microvascular dysfunction is also reported in patients with HFpEF without history of obstructive coronary artery disease.[86] Microvascular dysfunction might be an early but important stage in the pathogenesis of HFpEF, and may be equally important for both ventricles.

Non-cardiac co-morbidities

Several non-cardiac co-morbidities such as obesity, diabetes mellitus, renal dysfunction, COPD and hypertension are known to adversely impact myocardial function and remodelling via systemic pathways, including inflammation and endothelial dysfunction. The adverse impact of these processes in HFpEF has traditionally been focused on the LV.[87] However, given that these involve systemically circulating factors, there may plausibly be simultaneous involvement of the RV, resulting in RV remodelling and dysfunction. Co-existence of LV and RV remodelling in early HFpEF has been described by Borlaug *et al.*[32]

In subjects without cardiovascular disease, obesity may also be related to changes in RV structure and function.[88,89] Among the different co-morbidities, obesity has recently been shown to be associated with more profound abnormalities in RV–PA coupling.[58] As compared to HFpEF patients without obesity, patients with obesity display more RV remodelling, greater plasma volume expansion, more ventricular interdependence and pericardial restraint, higher filling pressures, and greater pulmonary vascular dysfunction during exercise. Right ventricular dilatation and dysfunction in obesity may be related in part to excessive volume loading, which is typical of obesity-related HFpEF.[58]

Several HFpEF-associated co-morbidities such as COPD may contribute to increased RV afterload via direct pulmonary vascular effects (Supplementary material online, *Figure S2*),[90] or impact RV remodelling by load-independent pathways.[91] Long-standing systemic hypertension leads to LV hypertrophy and stiffening, but may also result in RV remodelling, independent of pulmonary pressures.[92,93]

Furthermore, systemic hypertension often interacts with obesity and diabetes mellitus in the development of RV remodelling and dysfunction in patients with the metabolic syndrome, with early signs of RV diastolic dysfunction.[94–96]

Renal dysfunction and HFpEF often co-exist, and each condition may contribute to progression of the other in a vicious cycle via inflammation and endothelial dysfunction.[97] In left-sided heart failure, RVD is also strongly associated with renal dysfunction, mainly due to venous congestion resulting from chronic backward failure.[98] In a large cohort of 299 patients with HFpEF, renal dysfunction was reported to be independently associated with

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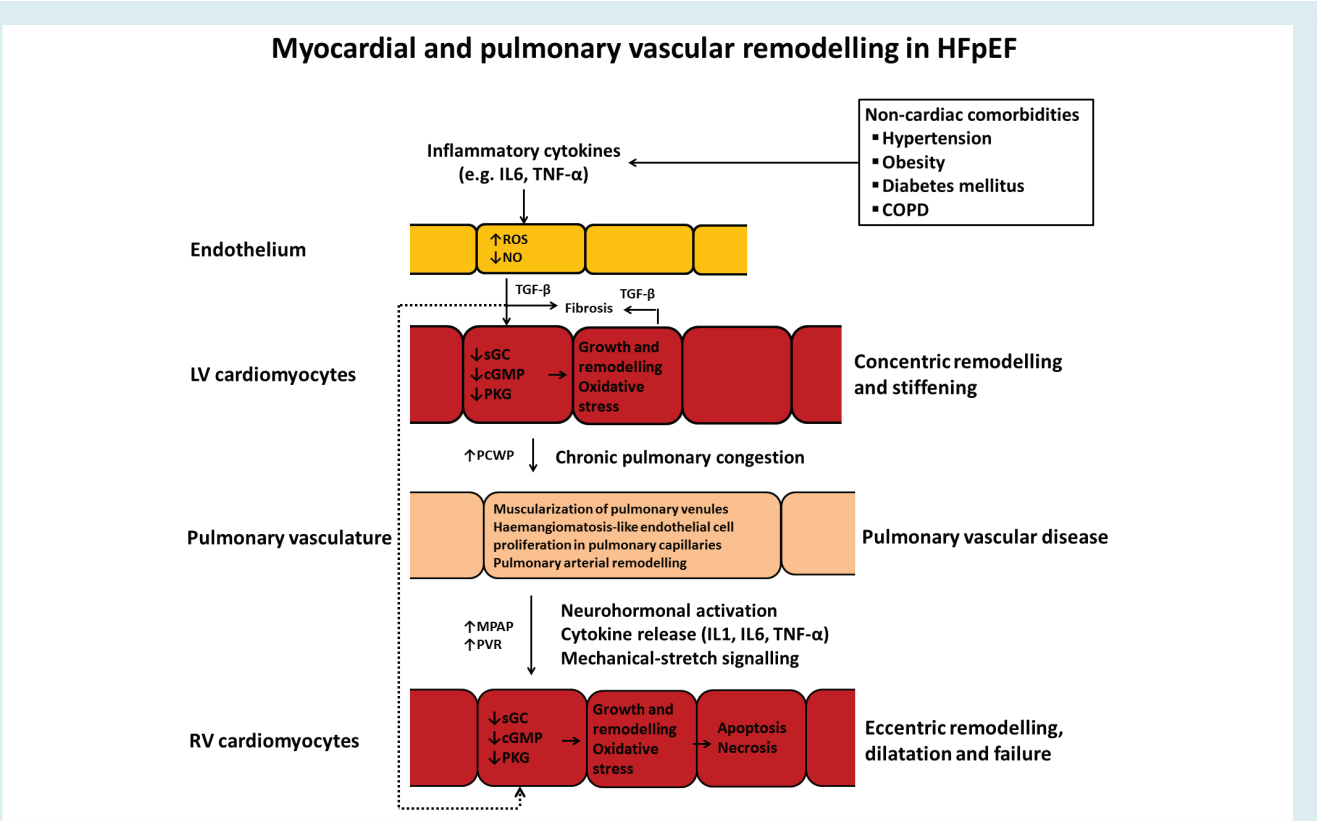


Figure 3 Pathophysiology of myocardial and pulmonary vascular remodelling in heart failure with preserved ejection fraction (HFpEF). Co-morbidities induce a systemic pro-inflammatory state with elevation of plasma cytokines, such as interleukin (IL)-6 and tumour necrosis factor (TNF)-α. Microvascular endothelial cells of the coronary vessels produce reactive oxygen species (ROS), which reduces the bioavailability of nitric oxide (NO) and this lowers the activity of soluble guanylate cyclase (sGC) in the cardiomyocytes. The resulting cascade of lower cyclic guanosine monophosphate (cGMP) and protein kinase G (PKG) ultimately results in myocardial remodelling and release of transforming growth factor (TGF)-β with deposition of collagen in the extracellular space. In theory, both ventricles are equally prone to these systemic mechanisms. However, the left ventricle (LV) displays more concentric remodelling, myocardial stiffening and diastolic dysfunction due to collagen deposition and fibrosis. On the contrary, the right ventricle (RV) exhibits more eccentric remodelling with hypertrophy, dilatation and systolic failure, mainly due to concomitant pulmonary vascular disease and resulting increased RV afterload. Chronic pulmonary congestion in the setting of increased LV filling pressures in HFpEF leads to remodelling changes of the pulmonary vasculature. The resulting increase in pulmonary artery pressures and RV overload triggers neurohormonal activation and cytokine release, which further activates the cGMP pathway in RV cardiomyocytes, leading to myocyte apoptosis and necrosis and ultimately RV failure. COPD, chronic obstructive pulmonary disease; PCWP, pulmonary capillary wedge pressure; PVR, pulmonary vascular resistance; MPAP, mean pulmonary artery pressure.

higher PA pressures and lower RV strain.[15] In addition, renal dysfunction strongly facilitates hypervolaemia, which is a major factor of right-sided decompensation, even in mild stages of right heart dysfunction.[99]

In some patients, wild-type transthyretin amyloidosis (ATTRwt) is an unrecognized cause of HFpEF.[100] The RV may also be involved in advanced ATTRwt (Supplementary material online, *Figure S3*), although its clinical relevance is currently unknown.

Pathophysiology

In left-sided heart failure, the RV myocardium may be subject to several stressors, where pressure overload in the setting of PH is most prominent. Chronic pulmonary congestion in left-sided heart failure leads to morphological changes of the pulmonary vasculature, including muscularization of pulmonary venules, haemangiomatosis-like endothelial cell proliferation in pulmonary capillaries and pulmonary arterial remodelling with intimal hypertrophy.[62] Pulmonary vascular remodelling in left-sided heart failure is different and seems more reversible than the remodelling patterns seen in patients with idiopathic PAH, in which there are more irreversible neointimal lesions such as concentric laminar intimal fibrosis and plexiform lesions.[101] The reversibility of severe PVD in patients with left-sided heart failure was previously demonstrated in a small group of patients with end-stage left-sided heart failure and presumably fixed high PVR. After implantation of LV assist device support, a progressive decrease in PVR and normalization of pulmonary pressures was observed.[102]

When the RV myocardium is exposed to increased afterload, several neurohormonal and molecular pathways are activated,

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such as cytokine release, activation of the endothelin system, the renin–angiotensin–aldosterone system (RAAS), the autonomic nervous system, and release of natriuretic peptides (*Figure 3*).[2] The myocardial wall of the normally unstressed RV thickens in order to maintain cardiac output. This remodelling pattern eventually leads to an increased mismatch between myocardial blood supply and oxygen demand, resulting in myocardial ischaemia and downstream effects such as collagen formation and fibrosis.[103] Oxidative stress triggers the production of reactive oxygen species, limits the availability of nitric oxide (NO) and significantly contributes to cell necrosis and apoptosis through release of inflammatory cytokines, such as tumour necrosis factor-α, interleukin-1, and interleukin-6. Oxidative stress and cytokine release degrade the extracellular matrix and myofibrils, and enhance collage formation, resulting in RV dilatation and myocardial fibrosis. The activation of the endothelin system is important in the setting of PVD and resultant RV failure.[2] Endothelin-1 is a potent vasoconstrictor and has pro-inflammatory and proliferative properties. Enhanced expression of endothelin-1 and endothelin receptors is seen in experimental PA banding models.[104] Elevated levels of catecholamines and overstimulation of the β-adrenergic receptors in the setting of RV failure, further stimulates maladaptive myocardial remodelling. For instance, in the presence of RV failure due to PA banding, decreased expression of β-adrenergic receptors in the RV was observed.[105]

Besides these cellular and molecular pathways involved in response to increased afterload, the changes in the pulmonary vasculature itself also lead to impaired pulmonary function and gas exchange.[62] Reduced diffusion capacity of the lung in patients with PVD-HFpEF is strongly associated with exercise intolerance and increased mortality.[106,107]

Finally, several of these pathways may also be activated in HFpEF—independent of PVD—and may be due to direct effects of co-morbidities such as obesity, hypertension and diabetes mellitus, which activate the release of reactive oxygen species via inflammatory cytokines and induce endothelial dysfunction and adverse remodelling of cardiomyocytes.[87] However, in patients with HFpEF, the LV displays more concentric myocardial remodelling, stiffening and diastolic dysfunction, while the RV shows more eccentric remodelling with hypertrophy, dilatation and systolic failure as a result of predominant increased afterload (*Figure 3*).

Treatment

The high prevalence of RVD and its potent prognostic consequences in HFpEF support the development of treatment strategies targeting the right side in HFpEF. The primary strategies of interest are (i) the reduction of pulmonary pressures in patients with PH-HFpEF using diuretics for congestion, and potentially novel vasoactive drugs targeting the cyclic guanosine monophosphate (cGMP) and endothelin pathways, and (ii) to directly target RV myocardial tissue. Until now, many previous attempts have demonstrated neutral results.[62] There is even concern that several PH-targeted therapies could rather have detrimental effects in HFpEF, due to rapid increases in LV filling pressures and resulting acute pulmonary oedema.[108] As a consequence, there are currently no established strategies to treat PVD and RVD in HFpEF, and the current European Society of Cardiology guidelines for the management of PH therefore provide a class III recommendation for PAH-approved treatment for patients with PH due to left heart disease (group 2 PH).[52] Potential treatment strategies are summarized in *Table 3*. The majority of these therapies target the different pathways described above.

General considerations

A cornerstone in the current management of symptomatic patients with HFpEF, particularly those with signs of right heart failure, is adequate volume management. Excessive volume overload is a major driver of decompensation, deterioration of RV function and multi-organ dysfunction (e.g. hepatic, renal, and splanchnic dysfunction). In clinical practice, in many patients with RVD-HFpEF diuretics are not given in sufficiently high dosages to ensure decongestion/euvolaemia, and early identification of patients who are at risk for diuretic resistance is important.[109,110] In addition, patients should be counselled to avoid excessive fluid and salt intake.

Furthermore, a multidisciplinary strategy with cooperation of a pulmonologist may be recommended in relevant cases, to rule out pulmonary causes of PH. Of note, drugs approved for PAH are currently also not recommended in patients with PH due to lung diseases (class IIIC recommendation).[52]

Besides anticoagulation, there are currently no specific treatments that improve prognosis in patients with HFpEF and AF.[71] Rate control is suggested in elderly, symptomatic patients to reduce symptomatic burden, improve ventricular filling time and prevent decongestion.[71] Rhythm control in patients with HFpEF, especially in an advanced stage with older age and multi-morbidity, is rather challenging, yet ongoing studies with early rhythm control strategies in HFpEF may hold promise in this regard.[71]

Obese, older patients with stable HFpEF benefit from caloric restriction and aerobic exercise training.[111] It is recommended to encourage regular aerobic exercise training in all patients with heart failure, to improve their functional capacity and symptoms (class IA recommendation).[34]

Cyclic guanosine monophosphate and nitric oxide pathway

cGMP and its target protein kinase G (PKG) are regulators of ion channel conductance and cellular apoptosis, and have an effect on vascular smooth muscle relaxation, vasodilatation, and increased blood flow.[112] Phosphodiesterase (PDE) inhibitors prevent the breakdown of cGMP and thereby enhance the vasodilator effects of cGMP.[112]

The PDE type 5 inhibitor sildenafil is an established drug for patients with World Health Organization Group 1 PAH.[52] Sildenafil has also been tested in HFpEF, yet with mixed results.[113–115] In a recent post-hoc analysis from a Dutch trial, conducted in 52 HFpEF patients with mean PA pressure ≥25 mmHg and PCWP ≥15 mmHg, sildenafil did not improve TAPSE and RV S' compared to placebo.[116] In a sub-analysis from the RELAX trial in HFpEF patients

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Table 3 Treatment targets for the right heart in heart failure with preserved ejection fraction

	PAH	HFpEF	Treatment effects in HFpEF	Under study	Rationale
Established treatments					
• None					
Potential treatments					
cGMP pathway					
• PDE5 (sildenafil)	++	+/–	No effect in isolated post-capillary PH, has some potential in selected patients with high PVR[113–115]	No	
• PDE9A	+/–	?	None	Pre-clinical	
• Riociguat	++	+	↑Stroke volume and ↑cardiac index and ↓RV end-diastolic area, but no change in PAP or PVR[121]	NCT02744339	PA vasodilatation, anti-proliferative, anti-fibrotic and anti-inflammatory properties
• Vericiguat	?	?	None	NCT01951638	PA vasodilatation
Nitric oxide pathway					
• Inhaled NO	+	+	↓PAP, ↑CO, ↓PVR in patients with high PVR[125–127]	NCT02742129	Acute PA vasodilatation
Endothelin pathway					
• Bosentan	++	–	Acute pulmonary oedema[108]	No	
• Macitentan	++	–	Data from MELODY-1 suggest more fluid retention with macitentan	NCT02070991	Reduction in hypertrophy and stiffening in animal model
Prostacyclin pathway					
• Epoprostenol	++	–	↑Mortality in HFrEF[131]	No	
• Iloprost	++	+/–	↓PAP and ↓PVR[132]	No	
• Treprostinil	++	?	None	NCT03037580	PA vasodilatation
Beta-blockade					
• Nebivolol	?	?	↑6MWD in HFpEF[135]	NCT02053246	β-adrenergic antagonist with vasodilator properties
• Carvedilol	+/–	?	None	No	
RAAS inhibitors					
• ACEi/ARBs	+/–	+/–	None	No	
• Spironolactone	+/–	+/–	No changes in RV function and pulmonary pressures[150]	No	
Other strategies					
• Ranolazine	+	+	↓PAP, no change in PVR, ↑6MWD[151,152]	No	
• PDE3 (milrinone)	+/–	?	None	No	

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Table 3 Continued

	PAH	HFpEF	Treatment effects in HFpEF	Under study	Rationale
• ARNI	?	+/−	No reduction in TR velocity[158]	No	
• CCBs	+	+/−	None	No	
• Digoxin	+/−	+/−	None	No	
• Apelin	+	?	None	No	
• Albuterol	+	+	Improved pulmonary vascular function[27]	NCT02885636	PA vasodilatation
• IASD	?	?	↓LA pressure, ↑RA pressure/volume	NCT01913613	Reduction in LA pressures
• PA denervation	?	?	None	NCT02220335	Destruction of pulmonary baroreceptors and sympathetic nervous fibres
• Wireless PAP monitoring	?	+	↓HF hospitalizations.[171,172] Class IIb-B recommendation in HF to reduce the risk of recurrent HF hospitalization[34]	No	
• Aerobic exercise training	++	++	Class IA recommendation to improve functional capacity and symptoms in HF[34]	NCT02435667 NCT02636439	Aerobic training improves exercise capacity and symptoms

(++) established beneficial effect; (+) non-established beneficial effect; (+/−) neutral effect; (−) harmful effect; (?) any effect unknown.
ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; CCB, calcium channel blocker; cGMP, cyclic guanosine monophosphate; CO, cardiac output; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; IASD, intra-atrial shunt device; LA, left atrium; 6MWD, 6-minute walking distance; NO, nitric oxide; PA, pulmonary artery; PAH, pulmonary arterial hypertension; PAP, pulmonary artery pressure; PDE, phosphodiesterase; PH, pulmonary hypertension; PVR, pulmonary vascular resistance; RA, right atrium; RAAS, renin–angiotensin–aldosterone system; RV, right ventricle; TR, tricuspid regurgitation.

with and without PH, sildenafil also did not improve TAPSE.[117] An earlier Italian trial, in patients with HFpEF and high PVR, did show improvement in pulmonary pressures, TAPSE and invasively assessed RV contractile state in the sildenafil treatment group.[113] The findings of a partial functional recovery in advanced RV failure associated with HFpEF by sildenafil were paralleled by a study providing mechanistic insights (i.e. restoration of cGMP pulmonary vascular concentration) on reported benefits. However, it remains to be evaluated whether carefully selected HFpEF patients with combined post- and pre-capillary PH will benefit from PDE type 5 inhibition.

Very recently, the PDE type 5A inhibitor vardenafil was tested in an animal model with Zucker diabetic fatty rats. It was reported that the development of HFpEF (defined as myocardial stiffness and diastolic dysfunction) was prevented and the activity of the cGMP–PKG axis was restored in this experimental diabetes mellitus-associated HFpEF model.[118] Whether early treatment with PDE type 5 inhibition may prevent worsening heart failure in humans with diabetes mellitus-associated HFpEF is unknown.[119]

PDE type 9A regulates natriuretic peptide-stimulated rather than NO-stimulated cGMP. PDE type 9A is expressed in human myocytes, is upregulated in the setting of myocardial hypertrophy and heart failure, and has been suggested as a potential therapeutic target in stress-induced heart disease.[120] Its potential as a target in patients with HFpEF and predominant RV failure is currently unknown.

Riociguat is a soluble guanylate-cyclase stimulator that induces vasodilatation, has anti-fibrotic, anti-proliferative, anti-inflammatory properties, and is effective in patients with PVR.[52] In the small DILATE-1 trial, riociguat did not improve mean PA pressure in patients with PH-HFpEF, but did increase stroke volume and cardiac index.[121] In addition, riociguat significantly decreased systolic blood pressure, systemic vascular resistance and RV end-diastolic area, without changing PCWP or PVR.[121] Riociguat is currently being tested in another phase 2 trial for its effect on pulmonary haemodynamics as well as RV size and function on CMR, in patients with PH-HFpEF (DYNAMIC, ClinicalTrials.gov NCT02744339).

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Another phase 2 trial with the soluble guanylate-cyclase stimulator vericiguat in HFpEF (SOCRATES-PRESERVED) has recently been completed. Vericiguat was well tolerated, did not change NT-proBNP and left atrial volume at 12 weeks compared to placebo, but was associated with improvement in quality of life.[122]

Recent data from the Swedish Heart Failure Registry in an unselected HFpEF population demonstrated that the use of conventional nitrates was not associated with improvements in all-cause mortality or heart failure hospitalizations.[123] In the NEAT-HFpEF randomized controlled trial, the patients with HFpEF who received isosorbide mononitrate were less active, and did not have better quality of life or submaximal exercise capacity compared to patients who received placebo.[124]

Inorganic nitrite administration represents a promising new therapy being tested in HFpEF. Nitrate and nitrite were formerly considered as inert by-products of NO metabolism, but it is now clear that these anions function as an important *in vivo* reservoir to generate NO under hypoxic circumstances in a NO synthase-independent fashion. Nitrite is reduced to NO by a variety of proteins including haemoglobin and myoglobin. This reaction is enhanced in the setting of hypoxia and acidosis, as develops in the tissues and veins during exercise. Intravenous nitrite was shown to reduce PA pressure and biventricular filling pressures in patients with HFpEF, with greater effect during exercise.[125] Cardiac output reserve was also enhanced. Inhaled, nebulized sodium nitrite has also been shown to reduce PA pressures and lower filling pressures at rest and during exercise, with improvements in PA compliance.[126,127] While acute administration of nitrite did not reduce PVR in the entire group of patients with HFpEF, it did lower PVR in those patients with higher PVR.[126] Nitrite is currently being tested in a National Institute of Health sponsored trial: Inorganic Nitrite Delivery to Improve Exercise Capacity in HFpEF (INDIE-HFpEF, ClinicalTrials.gov NCT02742129).

Endothelin pathway

Endothelin receptor antagonists such as bosentan are effective in World Health Organization Group 1 PAH.[52] On the contrary, bosentan exerts rather detrimental effects in patients with PH-HFpEF due to acute pulmonary oedema.[108]

Recently, it was demonstrated that endothelin-1 levels were elevated in humans with chronic stable HFpEF and these patients had increased myocardial hypertrophy.[128] In the same translational study, the endothelin-1 receptor antagonist macitentan reduced myocardial hypertrophy and stiffening via antihypertrophic mechanisms in a murine HFpEF model.[128] Endotelin-1 levels were also reported to be higher in HFpEF patients with diabetes mellitus compared to patients without diabetes mellitus.[129] Whether there is a link between endothelin-1 levels, diabetes mellitus and RV hypertrophy and dysfunction in HFpEF, remains unknown. The effects of macitentan on RV performance in HFpEF also require further study. Macitentan was recently under study in patients with combined post- and pre-capillary PH due to LV systolic and diastolic dysfunction (MELODY-1, ClinicalTrials.gov NCT02070991) and the primary results were recently presented at the 2017 Congress of the American College of Cardiology.[130] The macitentan group showed no change in PVR during 12 weeks. Moreover, this group was more likely to experience the primary composite endpoint of significant fluid retention or worsening functional class.[130]

Prostacyclin pathway

Prostacyclin is produced by endothelial cells and induces vasodilatation, inhibits platelet aggregation, and has anti-proliferative effects.[52] Prostacycline analogues such as epoprostenol and iloprost are established treatments for patients with PAH.[52] Chronic intravenous epoprostenol is associated with increased risk of mortality in patients with HFrEF, although the exact mechanisms are unclear.[131] In a small pilot study in patients with PH-HFpEF, acute inhalation of iloprost resulted in a significant reduction in PA pressure and PVR after 15 minutes.[132] No data regarding the effects of iloprost on RV performance in HFpEF are currently available. Treprostinil is another prostacyclin analogue that is indicated for the treatment of PAH,[52] and will be tested in patients with PH-HFpEF for its effect on 6-minute walk distance, functional class and NT-proBNP levels (ClinicalTrials.gov NCT03037580).

Neurohormonal inhibition

Chronic increased afterload of the RV activates the autonomic nervous system and the RAAS system. However, there is insufficient evidence whether neurohormonal inhibition is beneficial in the setting of RV failure.

Nebivolol is a β-adrenergic receptor antagonist with vasodilator properties. Nebivolol is not established in PAH. In HFpEF, the role of β-blockade is also uncertain. Data from the SENIORS trial suggest that nebivolol might be as useful in patients with HFpEF as it was for patients with HFrEF,[133] and observational data from the Swedish Heart Failure Registry also suggest that β-blockers may be beneficial in HFpEF.[134] However, the ELANDD trial did not show improvement by nebivolol in 6-minute walking distance in 116 patients with HFpEF, but no subgroup analyses for patients with RVD are available.[135] Hence, there is clearly insufficient evidence for the use of nebivolol in HFpEF to give it a place in the current guidelines, and more adequately powered studies are needed. Nebivolol is currently being tested in a phase 4 trial for its potential effect on PVR and 6-minute walk distance in patients with PH-HFpEF (ClinicalTrials.gov NCT02053246).

Carvedilol is a $\beta1/\beta2/\alpha1$-blocker, has anti-inflammatory and anti-oxidant properties, and might prevent pulmonary vascular remodelling.[136] Previously, carvedilol resulted in significant improvement in E/A ratio in patients with HFpEF from the SWEDIC trial.[137] Carvedilol has also been demonstrated to improve RV systolic function in patients with HFrEF,[138] yet any effect on the RV in HFpEF is currently unknown.

Several randomized controlled trials that investigated RAAS inhibition in HFpEF have not demonstrated improvement in outcome, and thus RAAS inhibition has no recommendation in the guidelines for the treatment of HFpEF.[34,139–141] However, observational data suggest that RAAS inhibition may be beneficial in heart failure with LV ejection fraction $\geq$40%.[142] RAAS inhibition was beneficial in several animal models with PVR, since it was reported to prevent RV hypertrophy, dysfunction and fibrosis, but published results

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are conflicting.[136,143,144] In an earlier trial, captopril was tested in 14 male patients with HFrEF in combination with reduced RV ejection fraction, and captopril was demonstrated to improve RV function and filling pressures.[145] Furthermore, it was also reported in another small study in 40 patients with systemic hypertension, that RAAS inhibition resulted in improvement in RV global function, independent of changes in systemic blood pressure.[146]

The TOPCAT trial investigated the aldosterone antagonist spironolactone in patients with HFpEF and demonstrated no reduction in the composite primary outcome of cardiovascular death, aborted cardiac arrest or hospitalization for heart failure.[147] However, spironolactone slightly reduced the risk of heart failure hospitalizations compared to placebo. In addition, subgroup analysis demonstrated that the primary outcome was met in the patients with elevated natriuretic peptides at baseline.[147] In the mechanistic Aldo-DHF trial in patients with HFpEF, spironolactone reduced E/e', an estimate of LV filling pressures, and led to improvements in cardiac structure (i.e. reduction in LV mass index) and reduction in NT-proBNP levels, but failed to improve symptoms or exercise capacity.[148] Aldosterone antagonists should be considered in the setting of fluid overload in patients with PAH.[52] It was reported in an animal model of PAH that spironolactone and eplerenone improved pulmonary haemodynamics.[149] In a sub-study from the TOPCAT trial in 239 patients with serial echocardiographic assessment, randomization to spironolactone was not associated with significant changes in RV function and pulmonary pressures.[150]

Based upon the observation that β-adrenergic stimulation improves pulmonary vascular function in HFpEF patients more than controls,[27] the Inhaled Beta-adrenergic Agonists to Treat Pulmonary Vascular Disease in Heart Failure With Preserved EF (BEAT-HFpEF, ClinicalTrials.gov NCT02885636) is a randomized controlled trial designed to investigate the inhalation of the β2-adrenergic agonist albuterol on changes in PVR in patients with HFpEF, with resting PCWP >15 mmHg or PCWP ≥25 mmHg with exercise.

Other drugs and experimental treatments

Ranolazine is an inhibitor of the late sodium channel current and is approved for the treatment of chronic stable angina. There is preliminary evidence from two small studies that it might have a potential in PAH. In one small open-label pilot study in 11 patients with symptomatic PAH, ranolazine (1000 mg twice daily) was administered for 3 months and was demonstrated to be safe and associated with improvement in functional class, reduction in RV size, and improvement in RV strain during exercise.[151] In another non-randomized pilot study in 10 patients with HFpEF and isolated post-capillary PH, mean PA pressure and PCWP significantly decreased, PVR remained unchanged, and 6-minute walking distance increased, during 6-month treatment with ranolazine (1000 mg daily).[152] Clearly, further evidence in adequately powered, prospective studies is needed to investigate the effects of ranolazine on prognostic endpoints in PH-HFpEF.

Milrinone is a PDE type 3 inhibitor that has been studied for reduction of RV afterload in patients undergoing LV assist device implantation,[153] in an experimental dog model with PA banding and RV failure,[154] and in infants with PH.[155] Recently, in a small study, Kaye *et al.*[156] investigated the use of intravenous milrinone in 20 patients with HFpEF and demonstrated improvement in pulmonary haemodynamics during exercise.

Recent recommendation suggests that levosimendan may be preferentially indicated over dobutamine in patients with PH caused by left heart disease.[99] Sacubitril/valsartan is an angiotensin receptor neprilysin inhibitor that has been proven to improve outcomes in patients with HFrEF.[157] Sacubitril/valsartan has previously also been tested in a phase 2 trial in 301 patients with HFpEF, but did not result in a reduction of TR velocity during 36 weeks of treatment, compared with placebo.[158]

Although the evidence is limited, calcium channel blockers may be effective for the treatment of systemic hypertension in HFpEF. Verapamil might have a beneficial effect on symptoms and diastolic dysfunction in patients with congestive heart failure and normal LV systolic function.[159] Calcium channel blockers are recommended in specific patients with PAH who demonstrate a favourable response to acute vasodilator testing, but close follow-up is warranted.[52]

Digoxin has no recommendation in the guidelines for the treatment of HFpEF.[34,160] There are also no convincing data available for the use of digoxin in the setting of PH.[52] In a small, non-randomized prospective study in 17 patients with PAH in combination with RV failure and normal LV function, treatment with digoxin modestly increased cardiac output.[161]

Apelin is an endogenous ligand for the APJ receptor that is present on endothelial cells, vascular smooth muscle cells, and cardiomyocytes.[162,163] It was previously reported from a small randomized trial in 20 patients with PAH that systemic infusion of apelin (10–100 nmol/min) resulted in increased cardiac output and stroke volume and in a reduction of PVR.[164] In two other small trials with respectively 8 and 12 subjects, administration of apelin infusion was also demonstrated to have a positive effect on cardiac output, systemic blood pressure and peripheral vascular resistance in patients with chronic left-sided heart failure.[163,165] Apelin agonism is currently not under investigation in patients with HFpEF.

Device therapies

The use of a transcatheter interatrial shunt device was safe and reduced left atrial pressures during exercise and thus demonstrated potential efficacy in patients with HFpEF.[166] This study excluded patients with right heart dysfunction and with evidence of PH with PVR >4 WU. During 6 months following the procedure, right atrial pressure and dimension, as well as RV end-diastolic volume, increased significantly.[166] Whether the observed haemodynamic effects, and the sudden change from pressure to volume overload of the right heart, will be beneficial during longer follow-up in patients with HFpEF remains to be evaluated. The interatrial shunt device is currently being studied in a randomized controlled, patient-blinded setting, with a

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Table 4 Knowledge gaps and future directions regarding the right side in heart failure with preserved ejection fraction

1	It is recommended to identify HFpEF patients with (additional) pulmonary vascular disease. More insight into the development of pulmonary vascular disease in HFpEF and adequate patient selection for future clinical trials is needed to develop and test more specific therapies that target the pulmonary vasculature in HFpEF.
2	There are several load-independent pathways for the development of RV dysfunction reported in cross-sectional studies, such as coronary artery disease and atrial fibrillation. However, clear cause–effect relations remain uncertain. Especially the hypothesis that the right heart is simultaneously affected in HFpEF, even in the early course of the disease, may suggest an intrinsic myocardial process and warrants further research.
3	There remains insufficient knowledge regarding right heart performance, and RV–pulmonary artery coupling and uncoupling, during exercise in HFpEF. More insight into these measures is warranted to address the previous research question as to what extent RV remodelling in HFpEF occurs simultaneously to left ventricular remodelling and independent of pulmonary pressures.
4	Obesity is highly prevalent in HFpEF and is associated with RV remodelling and dysfunction in the general community. More insight into the role of obesity and its effects on RV function and RV–pulmonary artery coupling is needed to investigate whether weight loss will be beneficial for right heart performance in HFpEF.
5	Endothelin-1 levels are associated with myocardial hypertrophy and stiffening in HFpEF. It has also been described that endothelin-1 levels are higher in HFpEF patients with diabetes mellitus than in patients without. It is recommended to investigate any link between endothelin, diabetes mellitus and RV hypertrophy and stiffening. Furthermore, whether PDE type 5 inhibition may be used as a preventive approach in patients with diabetes mellitus-associated HFpEF needs further study.
6	BNP is elevated in both left and right-sided overload and seems significantly more elevated in patients with HFpEF with additional pre-capillary pulmonary hypertension compared to HFpEF patients without pulmonary hypertension. There is an unmet need for a biomarker-signal profile unique to the right heart in patients with left-sided heart failure.
7	Many trials conducted in patients with HFpEF have not met the primary endpoints and thus these drugs have no place in the guidelines for the treatment of HFpEF. Some of these drugs have shown to be beneficial for the RV, both in experimental models and in the setting of pulmonary arterial hypertension. For the future, it might be recommended to investigate whether some of these drugs may have a positive effect in selected patients with HFpEF, when the endpoints of such trials are more focused on right-sided parameters.

BNP, B-type natriuretic peptide; HFpEF, heart failure with preserved ejection fraction; PDE, phosphodiesterase; RV, right ventricle.

non-implant control group (REDUCE LAP-HF, ClinicalTrials.gov NCT01913613).[167]

Daily monitoring of pulmonary pressures in patients with heart failure using a wireless PA pressure monitor—and by acting on elevation of pulmonary pressures with up-titration of diuretics, RAAS inhibitors or vasodilator drugs—significantly reduced PA pressures and hospitalizations for heart failure, both in trial setting and in real-world patients.[168–171] Wireless monitoring of PA pressures currently has a class IIb-B recommendation in order to reduce the risk of recurrent hospitalizations for heart failure.[34] In a sub-study from the CHAMPION trial in 119 patients with HFpEF, it was also shown that the number of hospitalizations for heart failure was reduced in the treatment group with active monitoring of pulmonary pressures compared with controls, and this reduction seemed to be primarily achieved by adequate titration of loop and thiazide diuretics.[170,172] Wireless monitoring of PA pressures may therefore be useful to control excessive volume overload in patients with HFpEF, to prevent recurrent episodes of acute decompensation and progression of multi-organ dysfunction. However, haemodynamic changes prior to an episode of acute decompensation may vary depending on the clinical presentation, and haemodynamic correlates of fluid retention may not be the same among individual patients and different situations.[173]

In another small study, pulmonary arterial denervation using a radiofrequency ablation catheter was reported to improve pulmonary haemodynamics in patients with PAH,[174,175] as well as in one patient with PH in combination with left-sided heart failure.[176] Pulmonary arterial denervation is currently being tested in patients with PH associated with left heart failure (PADN-5, ClinicalTrials.gov NCT02220335).

Conclusion

It is now well-established that RVD is highly prevalent and contributes to poor prognosis in HFpEF. We summarized available evidence regarding the prevalence, diagnosis, risk factors, pathophysiology, and potential therapeutic targets for the right heart in HFpEF. The present position paper also highlighted important remaining knowledge gaps and future directions (*Table 4*). Prospective studies are urgently needed to clarify the mechanisms underlying right heart remodelling and dysfunction, and to provide effective treatments that improve morbidity and mortality in HFpEF. We therefore advocate greater focus on the often neglected right side of the heart, and for the introduction of standardized endpoints of right heart dysfunction and failure in future clinical trials in HFpEF.

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Supplementary Information

Additional Supporting Information may be found in the online version of this article:

Figure S1. Inhomogeneous septal movement in a patient with mildly increased pulmonary pressures.

Figure S2. Example of a patient with combined heart failure with preserved ejection fraction and chronic obstructive pulmonary disease.

Figure S3. Right ventricular involvement in wild-type transthyretin amyloidosis.

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